9.4 $\underbrace{p \vee (q \to r)}_{f_1}, \underbrace{(p \vee q)}_{f_2}, \underbrace{\neg p}_{f_3} \vdash \underbrace{r}_{f_4}$

$f_1 = p \vee (q \to r) \equiv \neg p \to (q \to r)$

$f_2 = p \vee q \equiv \neg p \to q$

$f_2, f_3 \vdash_{mp} q \ (f_5)$

$f_1, f_3 \vdash_{mp} q \to r \ (f_6)$

$f_5, f_6 \vdash_{mp} r \ (f_7)$

$\vdash$ metasymbol: „f_4 derivable from f_3".

Inference rules:

$U, U \to V \vdash_{mp} V$

$\neg V, U \to V \vdash_{mt} \neg U$

$U \to V, V \to Z \vdash_{syllogism} U \to Z$

$U \to V \equiv \neg U \vee V$

$U \wedge V \vdash_{simplif} U$

Conclusion: The sequence of formulas $(f_1, f_2, f_3, f_5, f_6, f_4)$ is the deduction (proof) of $f_4 = r$ from the hypothesis f_1, f_2, f_3.

12.4.

H_1: It is not sunny (Su) this afternoon and it is colder (Co) than yesterday.

H_2: We will go swimming (Sw) only if it is sunny (Su).

H_3: If we do not go swimming, then we will take a canoe trip (Ct).

H_4: If we take a canoe trip, then we will be home by sunset (Hs).

C: We will go home by sunset.

Is C deducible from $\{H_1, H_2, H_3, H_4\}$?

$H_1, H_2, H_3, H_4 \overset{?}{\vdash} C$

$H_1: \neg Su \wedge Co$

$H_2: \neg Su \to \neg Sw$ (only if)

$H_3: \neg Sw \to Ct$

$H_4: Ct \to Hs$

$H_1 \vdash_{simp} \neg Su \ (f_5)$

$f_5, H_2 \vdash_{mp} \neg Sw \ (f_6)$

$H_3, H_4 \vdash \neg Sw \to Hs \ (f_7)$

$H_3, H_4 \vdash_{syll}$

$f6, f7 \vdash_{mp} H_S(C)$

The sequence containing (H, ~H4, f5, f6, f7 = C) is the deduction (proof) of C from the hypothesis.